

MARS INTERNSHIP PROGRAM

Internship Project Report

# **Land Degradation Prediction System**

*A Machine Learning Approach to Land Use Land Cover (LULC) based  
Land Degradation Classification across Uttar Pradesh*

Odd Semester, AY 2026–27

# 1. Introduction

Land is one of the most fundamental natural resources, supporting agriculture, biodiversity, and human settlement. Over the past few decades, rapid urbanization, deforestation, unsustainable agricultural practices, and climate variability have accelerated the degradation of land quality across many regions of India, including the state of Uttar Pradesh. Left unchecked, land degradation reduces agricultural productivity, disrupts local ecosystems, and increases vulnerability to droughts and soil erosion.

Traditional methods of assessing land degradation rely heavily on manual field surveys and visual interpretation of satellite imagery, which are time-consuming, subjective, and difficult to scale across large geographic areas. With the increasing availability of remote-sensing-derived environmental data, machine learning offers a faster and more consistent alternative for identifying and classifying degraded land at scale.

## Why LULC and Land Degradation Matter

- **Land Use Land Cover (LULC):** Classifies the earth's surface into categories — forest, agriculture, built-up, water, and barren land — using satellite-derived indicators.
- **Land Degradation:** The decline in land quality caused by deforestation, urban expansion, soil erosion, and unsustainable land use over time.
- **Why it matters:** Identifying degraded land early helps prioritize restoration and supports better land-use policy decisions.
- **My approach:** Instead of relying on manual visual inspection, this project uses a trained machine learning model to automatically classify degradation severity from environmental data.

## Problem Statement

Given land cover fractions and climate/soil indicators for a grid cell in Uttar Pradesh, the goal is to predict the severity of land degradation — Low, Moderate, or High — so that restoration efforts can be prioritized efficiently and objectively, rather than relying on ad-hoc manual assessment.

## Study Area: Uttar Pradesh

| Attribute          | Detail                                                       |
|--------------------|--------------------------------------------------------------|
| State              | Uttar Pradesh, India                                         |
| Districts covered  | 20 (including Chandauli, Badaun, Banda, Aligarh, and others) |
| Grid resolution    | 5 km × 5 km                                                  |
| Time period        | 2020–2024                                                    |
| Total observations | 22,645 grid cells                                            |

*A wide mix of cropland, forest, grassland, and built-up land is represented across the 20 districts, along with diverse hydrology and soil-moisture conditions across the Gangetic plain. This state-wide*

*grid gives the model enough variety in land cover and climate to generalize degradation patterns across districts.*

## 2. Objective

The primary objective of this project is to build and deploy a machine-learning system that can automatically classify land degradation severity across Uttar Pradesh. Specific objectives include:

1. To aggregate land cover, climate, and soil data into a uniform  $5 \text{ km} \times 5 \text{ km}$  grid across 20 districts of Uttar Pradesh.
2. To design a Land Degradation Index (LDI) that quantifies degradation severity from environmental indicators.
3. To train and tune a classification model capable of predicting degradation severity (Low, Moderate, High) with high confidence.
4. To identify which environmental features most strongly influence degradation, using explainability techniques (SHAP).
5. To deploy the model as an accessible platform supporting single-cell prediction, batch prediction, EDA, and geographic visualization for real-world use by authorities and communities.

### 3. Methodology

#### Tools & Technologies Used

| Layer                | Technology                                                      |
|----------------------|-----------------------------------------------------------------|
| Programming Language | Python                                                          |
| Machine Learning     | scikit-learn (Logistic Regression, preprocessing, model tuning) |
| Explainability       | SHAP (SHapley Additive exPlanations)                            |
| Backend / Deployment | Flask                                                           |
| Data Handling        | Pandas, NumPy                                                   |
| Visualization        | Matplotlib / Plotly (EDA and geographic dashboards)             |

#### Data Sources

| Category                 | Details                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------|
| Land Cover Fractions     | Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland (8 classes per grid cell) |
| Climate & Soil Variables | NDVI, Rainfall, Temperature, Soil Moisture (averaged per grid cell)                                    |

#### Workflow

1. Aggregate land cover fractions and climate/soil variables into 5 km × 5 km grid cells.
2. Compute a Land Degradation Index (LDI) per cell from these environmental indicators.
3. Label each cell's LDI into Low / Moderate / High degradation classes.
4. Train and tune a Logistic Regression classifier (13 environmental + 19 district features).
5. Validate performance and interpret feature influence using SHAP explainability.

#### Features Used in the Model

13 environmental variables and district identity together feed the classifier, totaling 32 input features per grid cell.

- **Land Cover Fractions (8):** Cropland, TreeCover, Grassland, BareLand, Builtup, Water, Wetland, Shrubland.
- **Climate & Soil Variables (5):** NDVI\_mean, Rainfall\_mean, Temperature\_mean, SoilMoisture\_mean, Area\_km2.
- **District Identity (19):** Each of the 20 Uttar Pradesh districts is one-hot encoded, letting the model capture district-specific baseline conditions.

**Total input features per grid cell: 13 environmental + 19 district (one-hot) = 32**

### Degradation Severity Classification

Every grid cell is assigned one of three degradation classes:

| Class    | Description                                                                                  |
|----------|----------------------------------------------------------------------------------------------|
| Low      | Healthy vegetation cover, stable soil moisture, minimal bare land or built-up encroachment.  |
| Moderate | Noticeable vegetation loss or rising bare/built-up share; land under stress but recoverable. |
| High     | Severe vegetation loss, high bare land, low soil moisture — urgent restoration needed.       |

### Model Selected

A Tuned Logistic Regression model was selected for deployment, trained on 13 environmental features plus district identity. It outputs the Low / Moderate / High class with a confidence score for every 5 km × 5 km grid cell.

| Attribute | Detail                                   |
|-----------|------------------------------------------|
| Model     | Tuned Logistic Regression                |
| Task      | Multiclass classification                |
| Classes   | Low, Moderate, High                      |
| Features  | 13 environmental + 19 district (one-hot) |

## 4. Result

### Deployed Platform

The trained model was deployed as an interactive platform (Flask backend) with six modules covering prediction, analysis, and interpretation:

- **Single Prediction** — Enter values for one grid cell and get an instant Low/Moderate/High prediction with confidence score.
- **Batch Prediction** — Upload a CSV of many grid cells at once; tested successfully on 22,645 rows in one run.
- **Model Performance** — Metrics and validation results for the Tuned Logistic Regression classifier.
- **EDA Dashboard** — Explore observations by year and class, including distributions, correlations, and scatter analysis.
- **SHAP Explainability** — Shows which features (temperature, rainfall, soil moisture, etc.) drive each prediction.
- **Geographic Dashboard** — Maps degradation results spatially across Uttar Pradesh's districts.

### Testing & Validation

The trained model and preprocessing artefacts (best\_model, preprocessor\_lr, label\_encoder) were validated on the full dataset of 22,645 grid-cell observations. Batch prediction was tested end-to-end by running all 22,645 rows through the deployed pipeline in a single execution, confirming that the system handles state-wide scale reliably. Individual predictions, such as the Chandauli example below, were cross-checked against expected degradation patterns based on known land cover composition.

### Sample Prediction — Chandauli District

| Field                   | Value     |
|-------------------------|-----------|
| District                | Chandauli |
| Area (km <sup>2</sup> ) | 20.42     |
| Cropland                | 73.84%    |
| TreeCover               | 7.45%     |
| BareLand                | 0.23%     |
| Built-up                | 2.98%     |
| NDVI (mean)             | 0.475     |
| Rainfall (mean)         | 1051.5 mm |
| Temperature (mean)      | 25.44 °C  |
| Soil Moisture (mean)    | 0.240     |

**Predicted Class: Moderate (Confidence: 100.0%, Model: Tuned Logistic Regression)**

| Class    | Probability |
|----------|-------------|
| Low      | 0.0%        |
| Moderate | 100.0%      |
| High     | 0.0%        |

*This grid cell shows strong cropland dominance with moderate tree cover — flagged for monitoring rather than immediate intervention.*

**Scope Achieved**

20 districts covered · 5 km × 5 km grid resolution · 22,645 grid observations · 3 severity classes · all model artefacts (best\_model, preprocessor\_lr, label\_encoder) deployed and ready.



## 5. Conclusion

This project successfully demonstrates that a machine-learning-based approach can classify land degradation severity efficiently and consistently across a large geographic area, removing the need for slow, manual visual assessment.

- A Tuned Logistic Regression model successfully classifies land degradation (Low/Moderate/High) across 22,645 grid cells in 20 Uttar Pradesh districts.
- The live platform offers single and batch prediction, EDA, model performance evaluation, and SHAP explainability — all deployed and working.

## Limitations

- The model currently relies on aggregated climate and land-cover indicators rather than raw high-resolution satellite imagery, which may limit fine-grained accuracy at sub-grid scale.
- Coverage is presently restricted to Uttar Pradesh and the 2020–2024 period; generalization to other states or time periods has not yet been validated.

## Future Scope

- Extend coverage to additional states beyond Uttar Pradesh.
- Automate satellite data ingestion to reduce manual data preparation.
- Add mobile access for field officers to enable on-ground verification and reporting.
- **Goal:** support local authorities and communities in prioritizing restoration efforts using data-driven insights.

## 6. Reference

The following AI tools and platforms were used during the development of the Land Degradation Prediction System, primarily for building and refining the web platform:

- **Claude AI** — used for code assistance and development support.
- **Google Antigravity** — used for platform/UI development assistance.
- **ChatGPT** — used for code assistance and development support.
- **Stitch AI** — used for UI/design assistance.

Additional technical references:

- Scikit-learn Documentation — Logistic Regression, Model Tuning and Evaluation. <https://scikit-learn.org/>
- SHAP (SHapley Additive exPlanations) Documentation — Model Explainability. <https://shap.readthedocs.io/>
- Flask Documentation — Web Application Backend. <https://flask.palletsprojects.com/>